Chronic dietary risk characterization for pesticide residues: a ranking and scoring method integrating agricultural uses and food contamination data.

A method has been developed to identify pesticide residues and foodstuffs for inclusion in national monitoring programs with different priority levels. It combines two chronic dietary intake indicators: ATMDI based on maximum residue levels and agricultural uses, and EDI on food contamination data. The mean and 95th percentile of exposure were calculated for 490 substances using individual and national consumption data. The results show that mean ATMDI exceeds the acceptable daily intake (ADI) for 10% of the pesticides, and the mean upper-bound EDI is above the ADI for 1.8% of substances. A seven-level risk scale is presented for substances already analyzed in food in France and substances not currently sought. Of 336 substances analyzed, 70 pesticides of concern (levels 2-5) should be particularly monitored, 22 of which are priority pesticides (levels 4 and 5). Of 154 substances not sought, 36 pesticides of concern (levels 2-4) should be included in monitoring programs, including 8 priority pesticides (level 4). In order to refine exposure assessment, analytical improvements and developments are needed to lower the analytical limits for priority pesticide/commodity combinations. Developed nationally, this method could be applied at different geographic scales. Copyright © 2011 Elsevier Ltd. All rights reserved.